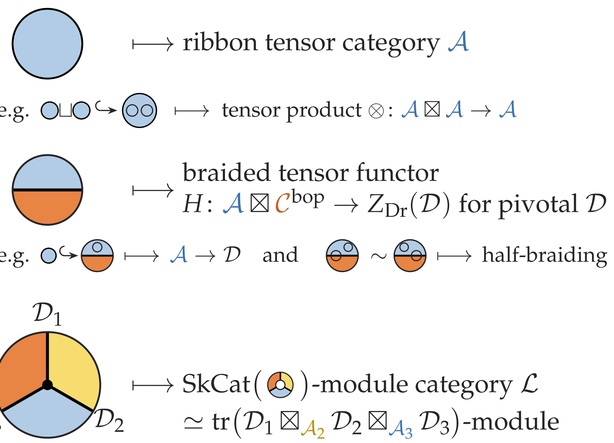
From invertibility of Borel defects to weaves and clusters

Matthias Vancraeynest
University of Edinburgh
matthiasvancraeynest.github.io

Defect skein theory

Algebraic data

Principle
Governed by disk embeddings and isotopies.

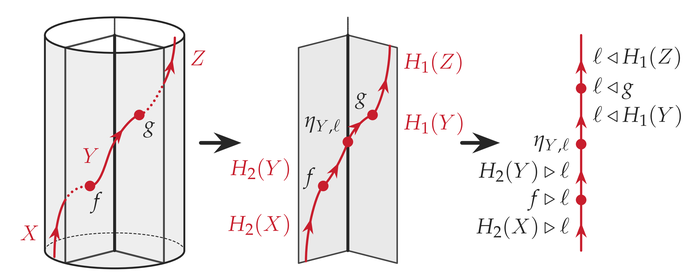


Evaluation functor

Theorem [Brown, Gómez García, Jordan, V. '26]
The following defines a full linear functor from decorated ribbon graphs to $\mathcal{L}$:

$$\mathrm{RT}_{\mathcal{L}}\colon \mathrm{Rib}_{\mathcal{L}}\longrightarrow\mathcal{L}.$$

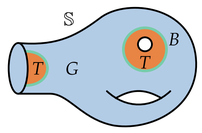
Skein relations: $\Gamma_1\sim\Gamma_2 \iff \mathrm{RT}(\Gamma_1)=\mathrm{RT}(\Gamma_2)$.



1. Isotope to a generic representative.
2. Project and relabel.
3. Isotope to separate vertices.
4. Evaluate in each sector.

Composing Borel defects

Quantum decorated character stacks [JLSS21]



$$\rightsquigarrow\ \mathrm{Ch}_G(\mathbb{S})=\left\{\begin{array}{l}G\text{-local systems on }G\text{-regions;}\\ T\text{-local systems on }T\text{-regions;}\\ B\text{-reductions on }B\text{-lines}\end{array}\right\}/\sim$$

$\rightsquigarrow$ recover moduli spaces $\mathcal{A}_{G,S}$, $\mathcal{P}_{G,S}$ of Fock-Goncharov and G-Shen.

$T\longmapsto \mathrm{Rep}_q T$, $G\longmapsto \mathrm{Rep}_q G$, $B\longmapsto \mathrm{Rep}_q B$ via $T\twoheadleftarrow B\hookrightarrow G$

For SL_2, they recover quantum cluster charts as subcategories, one for each triangulation Δ:

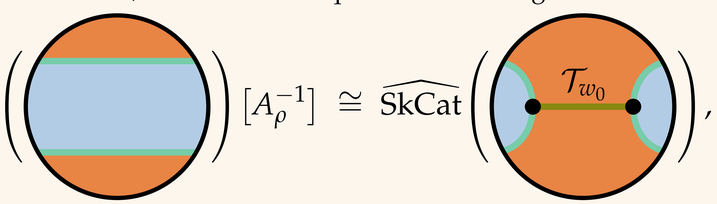
$$\mathrm{SkAlg}^{\mathrm{int}}(\mathbb{S})[X_\Delta^{-1}]\text{-mod}_{T^k}\hookrightarrow\widehat{\mathrm{SkCat}}(\mathbb{S}).$$

Generic invertibility

Borel defects are invertible after localisation, up to a Weyl symmetry.

Main theorem [Brown, Gómez García, Jordan, V. '26]
For any reductive G, there exists an equivalence of categories

$$\widehat{\mathrm{SkCat}}\left(\ \right)[A_\rho^{-1}]\ \cong\ \widehat{\mathrm{SkCat}}\left(\ \right),$$

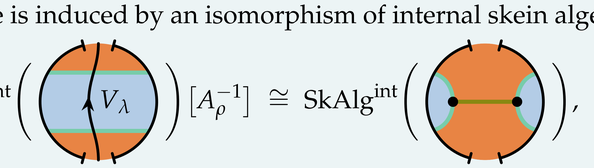


where A_ρ is the skein joining the two T-gates as below, with $\rho=\sum_i\varpi_i$.

Weyl defect: $\mathcal{T}_w\longmapsto \mathrm{Rep}_q T\boxtimes\mathrm{Rep}_q T^{\mathrm{bop}}\longrightarrow Z_{\mathrm{Dr}}(\mathrm{Rep}_q T)$, $\chi_\lambda\boxtimes\chi_\mu\longmapsto\chi_\lambda\otimes\chi_{w\cdot\mu}$

The equivalence is induced by an isomorphism of internal skein algebras:

$$\mathrm{SkAlg}^{\mathrm{int}}\left(\ \right)[A_\rho^{-1}]\ \cong\ \mathrm{SkAlg}^{\mathrm{int}}\left(\ \right),$$



and by monadic reconstruction

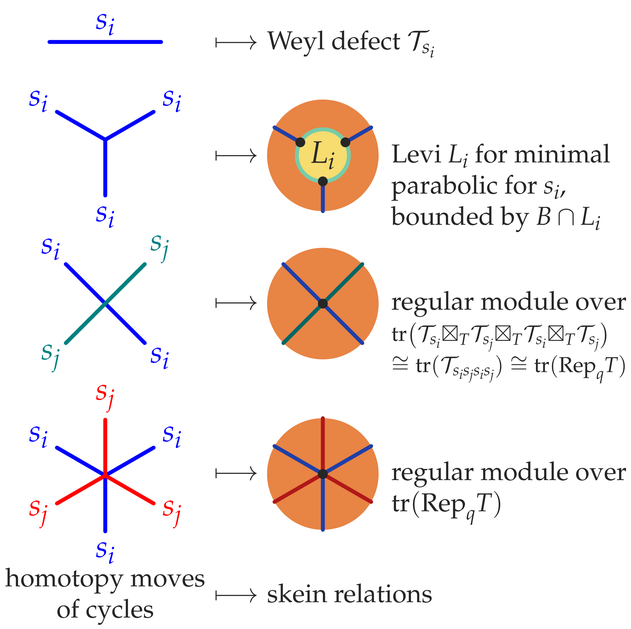
$$\widehat{\mathrm{SkCat}}(\mathbb{S})\longrightarrow\mathrm{SkAlg}^{\mathrm{int}}(\mathbb{S})\text{-mod}_{G^k\times T^l}.$$

Weaves as defects

Theorem (in progress)
A weave $\mathfrak{w}$ gives a decorated surface $\mathbb{S}_{\mathfrak{w}}$, and

$$\mathbb{W}(\mathfrak{w})\ \cong\ \mathrm{SkAlg}^{\mathrm{int}}(\mathbb{S}_{\mathfrak{w}}),$$

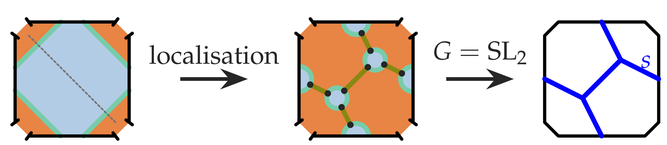
where $\mathbb{W}(\mathfrak{w})$ is the quantised weighted cycle algebra [Weng25].



Outlook
For general G, by extending the composition theorem to more general parabolics, we expect to describe

$$\mathbb{W}(\mathfrak{w}_\triangle)\text{-mod}_{T^k}\cong\widehat{\mathrm{SkCat}}(\mathbb{S})[X_\Delta^{-1}]\hookrightarrow\widehat{\mathrm{SkCat}}(\mathbb{S}),$$

identifying quantum cluster charts.



References
[BGJV26] Brown, Gómez García, Jordan, Vancraeynest, *Weyl defects in skein theory and quantum cluster charts*, arXiv:2609.09308 (2026)
[BJ25] Brown, Jordan, *Parabolic skein modules*, arXiv:2505.14836 (2025)
[JLSS21] Jordan, Le, Schrader, Shapiro, *Quantum decorated character stacks*, arXiv:2102.12283 (2021)
[Weng25] Weng, *Weighted cycles on weaves*, arXiv:2503.08020 (2025)

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